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COURSE CODE: SBT-DF203
LAB NAME: DNS SPOOFING FORENSICS
DATE: 23/09/2026
INSTRUCTOR: AMINU IDRIS

1. Source Materials and Slide Alignment

- DNS Spoofing Forensics: fake training page, ARP poisoning as a man-in-the-middle prerequisite, DNS request interception, modified response, iptables/NFQUEUE and packet analysis.
- The uploaded slide uses arp.py and dns_spoof.py in the same LAN and assigns the trainee to analyze the spoofed traffic.
- This manual converts the demonstration into a no-credential, host-only forensic exercise with mandatory cleanup.

Reading Requirement Before Lab Execution

The lab deliberately follows the sequence, terminology and core exercises presented in the uploaded SBT-DF203 slides. Commands have been corrected where needed for Linux syntax and forensic repeatability.

2. Lab Scenario

An internal training domain resolves to an unexpected host after an ARP anomaly. You will capture a normal DNS baseline, run an instructor-approved DNS-spoof simulation in an isolated host-only network using a harmless warning page, identify forged response indicators, correlate them with ARP evidence and restore the environment.

3. Learning Outcomes

- Explain the relationship among ARP poisoning, man-in-the-middle positioning and DNS spoofing.
- Capture baseline DNS and ARP evidence before a controlled simulation.
- Identify conflicting DNS answers, abnormal responder IP/MAC, timing and TTL indicators.
- Correlate a spoofed DNS response with the web connection that follows.
- Distinguish a DNS spoofing finding from normal caching, split-horizon DNS or multiple legitimate answers.
- Restore forwarding, iptables and ARP state after the lab.

4. Legal, Ethical and Safety Rules

- Use only systems, packet captures and networks for which you have explicit authorization.
- Keep the lab isolated from production, campus, office and public networks. Use NAT or host-only virtual networking as instructed.
- Do not collect, disclose or reuse real credentials, personal messages or confidential traffic.
- Preserve original capture files. Perform analysis on verified working copies and record SHA-256 hashes.
- Stop any traffic-generation or interception process immediately after the required evidence is captured.
- Restore firewall, ARP, forwarding and network settings before closing the lab.
- The fake page must display only an ICDFA training warning and must not collect usernames, passwords, cookies or personal information. Use a reserved training domain such as portal.icdfa.test or bank.training.invalid, never a real bank or public domain.

5. Required Tools and Environment

Item	Recommended Tool	Purpose
Virtual machines	Kali analyst plus victim VM	Run on a host-only/internal network.
Web server	Apache2 on analyst VM	Host harmless training page.
Capture tools	Wireshark/tshark	Capture DNS, ARP and HTTP.
Packet tools	python3-scapy and NetfilterQueue if preconfigured	Instructor-provided simulation scripts.
Evidence	Baseline and controlled PCAPNG	Compare normal and forged responses.
Test domain	Instructor-assigned .test/.invalid name	Avoid impersonating real organizations.

6. Lab Folder Structure and Evidence Preparation

```
mkdir -p ~/SBT-DF203-Lab8/{evidence,working,exported,reports,screenshots,scripts}
cd ~/SBT-DF203-Lab8
pwd
find . -maxdepth 1 -type d -print
```

Create the folder structure before downloading or generating evidence. Store original captures under evidence and analysis copies under working.

sudo apt update

sudo apt install -y apache2 tshark wireshark python3-scapy python3-pip build-essential python3-dev libnetfilter-queue-dev

sudo systemctl enable --now apache2

```
printf '<!DOCTYPE html>\n<html><body style="font-family:Arial"><h1>ICDFA DNS Spoofing Training
Page</h1><p>This is an authorized simulation. Do not enter credentials.</p><p>Analyst: YOUR
NAME</p></body></html>\n' | sudo tee /var/www/html/index.html
```

```
sha256sum /var/www/html/index.html | tee reports/training_page_sha256.txt
```

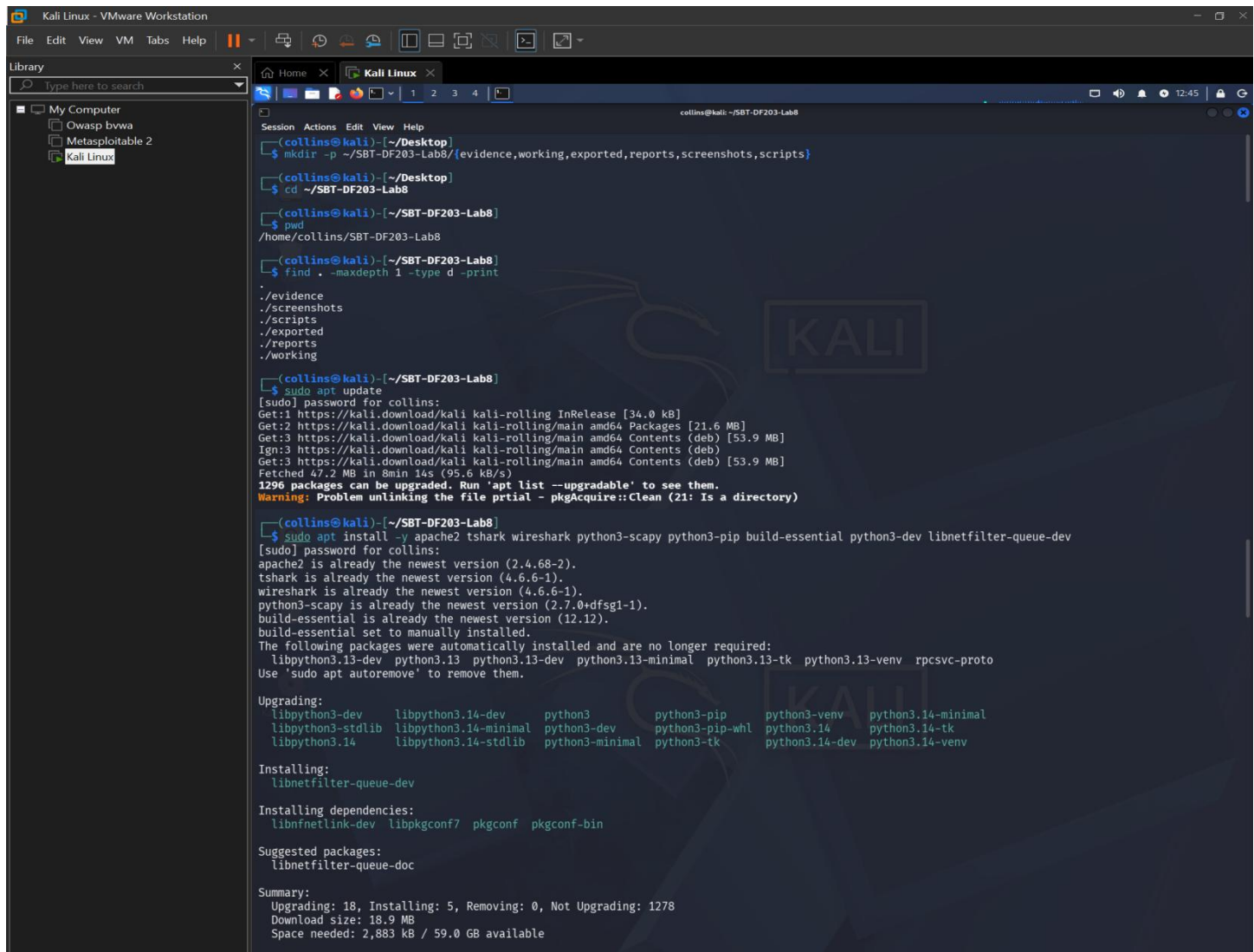
```
ip -br address | tee reports/interfaces.txt
```

```
ip route | tee reports/routes.txt
```

```
ip neigh show | tee reports/arp_before.txt
```

Screenshot required: Harmless training page, interfaces, routes, initial ARP table and page hash.

ANSWER:



```
Kali Linux - VMware Workstation
File Edit View VM Tabs Help
Library
My Computer
  Owasp bwba
  Metasploitable 2
  Kali Linux

(collins@kali) [~/Desktop]
$ mkdir -p ~/SBT-DF203-Lab8/{evidence,working,exported,reports,screenshots,scripts}

(collins@kali) [~/Desktop]
$ cd ~/SBT-DF203-Lab8

(collins@kali) [~/SBT-DF203-Lab8]
$ pwd
/home/collins/SBT-DF203-Lab8

(collins@kali) [~/SBT-DF203-Lab8]
$ find . -maxdepth 1 -type d -print
.
./evidence
./screenshots
./scripts
./exported
./reports
./working

(collins@kali) [~/SBT-DF203-Lab8]
$ sudo apt update
[sudo] password for collins:
Get:1 https://kali.download/kali kali-rolling InRelease [34.0 kB]
Get:2 https://kali.download/kali kali-rolling/main amd64 Packages [21.6 MB]
Get:3 https://kali.download/kali kali-rolling/main amd64 Contents (deb) [53.9 MB]
Ign:3 https://kali.download/kali kali-rolling/main amd64 Contents (deb)
Get:3 https://kali.download/kali kali-rolling/main amd64 Contents (deb) [53.9 MB]
Fetched 47.2 MB in 8min 14s (95.6 kB/s)
1296 packages can be upgraded. Run 'apt list --upgradable' to see them.
Warning: Problem unlinking the file partial - pkgAcquire::Clean (21: Is a directory)

(collins@kali) [~/SBT-DF203-Lab8]
$ sudo apt install -y apache2 tshark wireshark python3-scapy python3-pip build-essential python3-dev libnetfilter-queue-dev
[sudo] password for collins:
apache2 is already the newest version (2.4.68-2).
tshark is already the newest version (4.6.6-1).
wireshark is already the newest version (4.6.6-1).
python3-scapy is already the newest version (2.7.0+dfsg1-1).
build-essential is already the newest version (12.12).
build-essential set to manually installed.
The following packages were automatically installed and are no longer required:
  libpython3.13-dev python3.13 python3.13-dev python3.13-minimal python3.13-tk python3.13-venv rpcsvc-proto
Use 'sudo apt autoremove' to remove them.

Upgrading:
  libpython3-dev      libpython3.14-dev      python3      python3-pip      python3-venv      python3.14-minimal
  libpython3-stdlib  libpython3.14-minimal  python3-dev  python3-pip-whl  python3.14      python3.14-tk
  libpython3.14      libpython3.14-stdlib  python3-minimal  python3-tk      python3.14-dev  python3.14-venv

Installing:
  libnetfilter-queue-dev

Installing dependencies:
  libnftnl-dev libpkgconf7 pkgconf pkgconf-bin

Suggested packages:
  libnetfilter-queue-doc

Summary:
  Upgrading: 18, Installing: 5, Removing: 0, Not Upgrading: 1278
  Download size: 18.9 MB
  Space needed: 2,883 kB / 59.0 GB available
```

```

Get:1 https://http.kali.org/kali kali-rolling/main amd64 python3-pip-whl all 26.1.2+dfsg-1 [1,463 kB]
Get:2 https://kali.download/kali kali-rolling/main amd64 python3.14-venv amd64 3.14.7-3 [5,456 B]
Get:3 https://kali.download/kali kali-rolling/main amd64 python3.14-dev amd64 3.14.7-3 [508 kB]
Get:4 https://kali.download/kali kali-rolling/main amd64 libpython3.14-dev amd64 3.14.7-3 [5,820 kB]
Get:5 https://kali.download/kali kali-rolling/main amd64 libpython3.14 amd64 3.14.7-3 [2,530 kB]
Get:6 https://kali.download/kali kali-rolling/main amd64 python3.14 amd64 3.14.7-3 [861 kB]
Get:7 https://kali.download/kali kali-rolling/main amd64 python3.14-minimal amd64 3.14.7-3 [2,712 kB]
Get:8 https://kali.download/kali kali-rolling/main amd64 libpython3.14-minimal amd64 3.14.7-3 [902 kB]
Get:9 https://kali.download/kali kali-rolling/main amd64 libpython3.14-stdlib amd64 3.14.7-3 [2,360 kB]
Get:10 https://kali.download/kali kali-rolling/main amd64 libpython3-dev amd64 3.14.7-3 [8,492 B]
Get:11 https://kali.download/kali kali-rolling/main amd64 python3-dev amd64 3.14.7-3 [26.7 kB]
Get:12 https://kali.download/kali kali-rolling/main amd64 python3-venv amd64 3.14.7-3 [1,180 B]
Get:13 https://kali.download/kali kali-rolling/main amd64 python3-minimal amd64 3.14.7-3 [25.3 kB]
Get:14 https://kali.download/kali kali-rolling/main amd64 python3 amd64 3.14.7-3 [26.0 kB]
Get:15 https://kali.download/kali kali-rolling/main amd64 libpython3-stdlib amd64 3.14.7-3 [8,252 B]
Get:16 https://kali.download/kali kali-rolling/main amd64 libpkgconf7 amd64 2.5.1-4 [47.8 kB]
Get:17 https://kali.download/kali kali-rolling/main amd64 pkgconf-bin amd64 2.5.1-4 [35.9 kB]
Get:18 https://kali.download/kali kali-rolling/main amd64 pkgconf amd64 2.5.1-4 [33.6 kB]
Get:19 https://http.kali.org/kali kali-rolling/main amd64 libnfnetwork-dev amd64 1.0.2-3+b2 [8,220 B]
Get:20 https://http.kali.org/kali kali-rolling/main amd64 libnfnetwork-queue-dev amd64 1.0.5-4+b2 [18.5 kB]
Get:21 https://http.kali.org/kali kali-rolling/main amd64 python3-pip all 26.1.2+dfsg-1 [1,402 kB]
Get:22 https://kali.download/kali kali-rolling/main amd64 python3.14-tk amd64 3.14.7-3 [108 kB]
Get:23 https://kali.download/kali kali-rolling/main amd64 python3-tk amd64 3.14.7-3 [8,472 B]
Fetched 18.9 MB in 1min 51s (171 kB/s)
(Reading database... 426215 files and directories currently installed.)
Preparing to unpack ./00-python3-pip-whl_26.1.2+dfsg-1_all.deb...
Unpacking python3-pip-whl (26.1.2+dfsg-1) over (26.1.1+dfsg-1)...
Preparing to unpack ./01-python3.14-venv_3.14.7-3_amd64.deb...
Unpacking python3.14-venv (3.14.7-3) over (3.14.6-1)...
Preparing to unpack ./02-python3.14-dev_3.14.7-3_amd64.deb...
Unpacking python3.14-dev (3.14.7-3) over (3.14.6-1)...
Preparing to unpack ./03-libpython3.14-dev_3.14.7-3_amd64.deb...
Unpacking libpython3.14-dev:amd64 (3.14.7-3) over (3.14.6-1)...
Preparing to unpack ./04-libpython3.14_3.14.7-3_amd64.deb...
Unpacking libpython3.14:amd64 (3.14.7-3) over (3.14.6-1)...
Preparing to unpack ./05-python3.14_3.14.7-3_amd64.deb...
Unpacking python3.14 (3.14.7-3) over (3.14.6-1)...
Preparing to unpack ./06-python3.14-minimal_3.14.7-3_amd64.deb...
Unpacking python3.14-minimal (3.14.7-3) over (3.14.6-1)...
Preparing to unpack ./07-libpython3.14-minimal_3.14.7-3_amd64.deb...
Unpacking libpython3.14-minimal:amd64 (3.14.7-3) over (3.14.6-1)...
Preparing to unpack ./08-libpython3.14-stdlib_3.14.7-3_amd64.deb...
Unpacking libpython3.14-stdlib:amd64 (3.14.7-3) over (3.14.6-1)...
Preparing to unpack ./09-libpython3-dev_3.14.7-3_amd64.deb...
Unpacking libpython3-dev:amd64 (3.14.7-3) over (3.14.6-1)...
Preparing to unpack ./10-python3-dev_3.14.7-3_amd64.deb...
Unpacking python3-dev (3.14.7-3) over (3.14.6-1)...
Preparing to unpack ./11-python3-venv_3.14.7-3_amd64.deb...
Unpacking python3-venv (3.14.7-3) over (3.14.6-1)...
Setting up libpython3.14-minimal:amd64 (3.14.7-3)...
Setting up python3.14-minimal (3.14.7-3)...
(Reading database... 426217 files and directories currently installed.)
Preparing to unpack ./python3-minimal_3.14.7-3_amd64.deb...
Unpacking python3-minimal (3.14.7-3) over (3.14.6-1)...
Setting up python3-minimal (3.14.7-3)...
(Reading database... 426217 files and directories currently installed.)
Preparing to unpack ./0-python3_3.14.7-3_amd64.deb...
running python pre-rtupdate hooks for python3.14...
Unpacking python3 (3.14.7-3) over (3.14.6-1)...
Preparing to unpack ./1-libpython3-stdlib_3.14.7-3_amd64.deb...
Unpacking libpython3-stdlib:amd64 (3.14.7-3) over (3.14.6-1)...
Selecting previously unselected package libpkgconf7:amd64.
Preparing to unpack ./2-libpkgconf7_2.5.1-4_amd64.deb...
Unpacking libpkgconf7:amd64 (2.5.1-4)...
Selecting previously unselected package pkgconf-bin.
Preparing to unpack ./3-pkgconf-bin_2.5.1-4_amd64.deb...
Unpacking pkgconf-bin (2.5.1-4)...
Selecting previously unselected package pkgconf:amd64.
Preparing to unpack ./4-pkgconf_2.5.1-4_amd64.deb...
Unpacking pkgconf:amd64 (2.5.1-4)...
Selecting previously unselected package libnfnetwork-dev:amd64.
Preparing to unpack ./5-libnfnetwork-dev_1.0.2-3+b2_amd64.deb...
Unpacking libnfnetwork-dev:amd64 (1.0.2-3+b2)...
Selecting previously unselected package libnfnetwork-queue-dev:amd64.
Preparing to unpack ./6-libnfnetwork-queue-dev_1.0.5-4+b2_amd64.deb...
Unpacking libnfnetwork-queue-dev:amd64 (1.0.5-4+b2)...
Preparing to unpack ./7-python3-pip_26.1.2+dfsg-1_all.deb...
Unpacking python3-pip (26.1.2+dfsg-1) over (26.1.1+dfsg-1)...
Preparing to unpack ./8-python3.14-tk_3.14.7-3_amd64.deb...
Unpacking python3.14-tk (3.14.7-3) over (3.14.6-1)...
Preparing to unpack ./9-python3-tk_3.14.7-3_amd64.deb...
Unpacking python3-tk (3.14.7-3) over (3.14.6-1)...
Setting up python3.14-tk (3.14.7-3)...
Setting up python3-pip-whl (26.1.2+dfsg-1)...
Setting up libpython3.14-stdlib:amd64 (3.14.7-3)...
Setting up libpkgconf7:amd64 (2.5.1-4)...
Setting up pkgconf-bin (2.5.1-4)...
Setting up libpython3.14:amd64 (3.14.7-3)...
Setting up libpython3.14-dev:amd64 (3.14.7-3)...
Setting up python3.14 (3.14.7-3)...
Setting up libpython3-stdlib:amd64 (3.14.7-3)...
Setting up libpython3-dev:amd64 (3.14.7-3)...
Setting up python3.14-dev (3.14.7-3)...
Setting up python3-tk (3.14.7-3)...
Setting up python3 (3.14.7-3)...
running python rtupdate hooks for python3.14...
/usr/share/commix/src/core/tamper/backslashes.py:21: SyntaxWarning: '\' is an invalid escape sequence. Such sequences will not work in the future. Did you mean '\\'? A raw string is also an option.
  About: Adds back slashes (\) between the characters in a given payload.
/usr/share/fern-wifi-cracker/core/functions.py:100: SyntaxWarning: 'return' in a 'finally' block
  return read_file
running python post-rtupdate hooks for python3.14...
Setting up pkgconf:amd64 (2.5.1-4)...
Setting up python3.14-venv (3.14.7-3)...
Setting up python3-venv (3.14.7-3)...
Setting up python3-dev (3.14.7-3)...
Setting up python3-pip (26.1.2+dfsg-1)...
Setting up libnfnetwork-dev:amd64 (1.0.2-3+b2)...
Setting up libnfnetwork-queue-dev:amd64 (1.0.5-4+b2)...
Processing triggers for desktop-file-utils (0.28-1)...
Processing triggers for doc-base (0.11.2)...
Processing 1 changed doc-base file...
Processing triggers for libc-bin (2.43-4)...
Processing triggers for systemd (261.2-1)...
Processing triggers for man-db (2.13.1-1)...
Processing triggers for kali-menu (2026.2.6)...

(collins@kali) [~/SBT-DF203-Lab8]
$ sudo systemctl enable --now apache2

(collins@kali) [~/SBT-DF203-Lab8]
$ printf '<!DOCTYPE html><n<html><body style="font-family:Arial"><h1>ICDFA DNS Spoofing Training Page</h1><p>This is an authorized simulation. Do not enter credentials.</p><p>Analyst: COLLINS AMADI</p></body></html><n' | sudo tee /var/www/html/index.html
<!DOCTYPE html>
<html><body style="font-family:Arial"><h1>ICDFA DNS Spoofing Training Page</h1><p>This is an authorized simulation. Do not enter credentials.</p><p>Analyst: COLLINS AMADI</p></body></html>

```

```
(collins@kali) [~/SBT-DF203-Lab8]
$ sha256sum /var/www/html/index.html | tee reports/training_page_sha256.txt
7dd7121ccc1411d0d8940cd2bca282cfbbfa6aada262a5c8a716164ace66e618 /var/www/html/index.html

(collins@kali) [~/SBT-DF203-Lab8]
$ ip -br address | tee reports/interfaces.txt
lo        UNKNOWN    127.0.0.1/8 ::1/128
eth0      UP           192.168.188.130/24 fe80::20c:29ff:fec3:2528/64

(collins@kali) [~/SBT-DF203-Lab8]
$ ip route | tee reports/routes.txt
default via 192.168.188.2 dev eth0 proto dhcp src 192.168.188.130 metric 100
192.168.188.0/24 dev eth0 proto kernel scope link src 192.168.188.130 metric 100

(collins@kali) [~/SBT-DF203-Lab8]
$ ip neigh show | tee reports/arp_before.txt
192.168.188.1 dev eth0 lladdr 00:50:56:c0:00:08 STALE
192.168.188.254 dev eth0 lladdr 00:50:56:ed:5d:36 STALE
192.168.188.2 dev eth0 lladdr 00:50:56:e8:6a:f6 STALE

(collins@kali) [~/SBT-DF203-Lab8]
$
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

81°F
Light rain

Search

12:56 PM
9/22/2026

7. Mini Evidence and Chain-of-Custody Worksheet

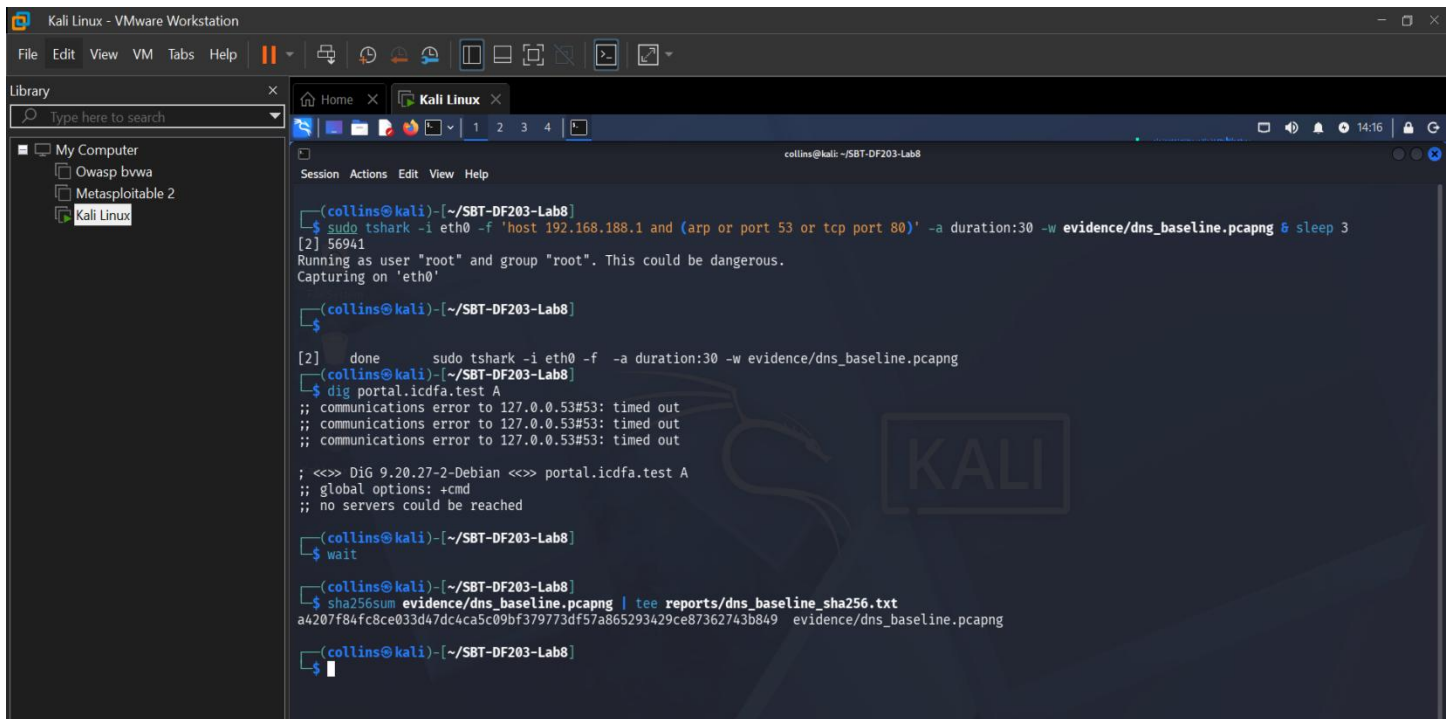
Field	Student Entry
Case/lab identifier	SBT-DF203-Lab8-COLLINS KELECHI AMADI
Trainee name	COLLINS AMADI
Date and time started	23-09-2026 12:53:45
Evidence file name(s)	/var/ww/html/index.html
Source or generation method	COLLINS@KALI
Original SHA-256	sha256sum /var/www/html/index.html tee reports/training_page_sha256.txt
Working-copy SHA-256	7dd7121ccc1411d0d8940cd2bca282cfbbfa6aada262a5c8a716164ace66e618
Analysis workstation/VM	KALI LINUX COMMAND
Notes on any changes	NILL

8. Part A - Document the Baseline

- 1. Record victim IP/MAC, analyst IP/MAC, gateway IP/MAC and DNS resolver IP.
- 2. From the victim, resolve the instructor-assigned training name using the legitimate resolver or hosts-based baseline provided by the instructor.
- 3. Capture ARP and DNS traffic before starting any spoofing process.
- 4. Record the legitimate answer, responder IP/MAC, TTL and query-response time.

```
# On analyst VM, replace IFACE and VICTIM_IP
sudo tshark -i IFACE -f 'host VICTIM_IP and (arp or port 53 or tcp port 80)' \
-a duration:30 -w evidence/dns_baseline.pcapng &
sleep 3
# On victim: dig portal.icdfa.test A (or instructor-provided name)
wait
sha256sum evidence/dns_baseline.pcapng | tee reports/dns_baseline_sha256.txt
```

ANSWER:



```
Kali Linux - VMware Workstation
File Edit View VM Tabs Help

Library
Type here to search
My Computer
  Owasp bwaa
  Metasploitable 2
  Kali Linux

Kali Linux
collins@kali: ~/SBT-DF203-Lab8

(collins@kali)~[~/SBT-DF203-Lab8]
$ sudo tshark -i eth0 -f 'host 192.168.188.1 and (arp or port 53 or tcp port 80)' -a duration:30 -w evidence/dns_baseline.pcapng & sleep 3
[2] 56941
Running as user "root" and group "root". This could be dangerous.
Capturing on 'eth0'

(collins@kali)~[~/SBT-DF203-Lab8]
$

[2] done      sudo tshark -i eth0 -f -a duration:30 -w evidence/dns_baseline.pcapng
(collins@kali)~[~/SBT-DF203-Lab8]
$ dig portal.icdfa.test A
;; communications error to 127.0.0.53#53: timed out
;; communications error to 127.0.0.53#53: timed out
;; communications error to 127.0.0.53#53: timed out

;<>>> DiG 9.20.27-2-Debian <<>> portal.icdfa.test A
;; global options: +cmd
;; no servers could be reached

(collins@kali)~[~/SBT-DF203-Lab8]
$ wait

(collins@kali)~[~/SBT-DF203-Lab8]
$ sha256sum evidence/dns_baseline.pcapng | tee reports/dns_baseline_sha256.txt
a4207f84fc8ce033d47dc4ca5c09bf379773df57a865293429ce87362743b849  evidence/dns_baseline.pcapng

(collins@kali)~[~/SBT-DF203-Lab8]
$
```

9. Part B - Prepare the Authorized Simulation

Instructor-Controlled Scripts

Use only the instructor-provided `arp.py` and `dns_spoof.py` from the course materials. Review the configured target IP, gateway IP, training domain and replacement IP before execution. The replacement IP must be the analyst VM hosting the warning page.

- Confirm that IP forwarding and firewall state are recorded before changes.
- Confirm the victim can reach the analyst training page directly by IP.
- Verify that the script matches only the reserved training domain.
- Set a 30-second timeout or stop manually as soon as evidence is captured.

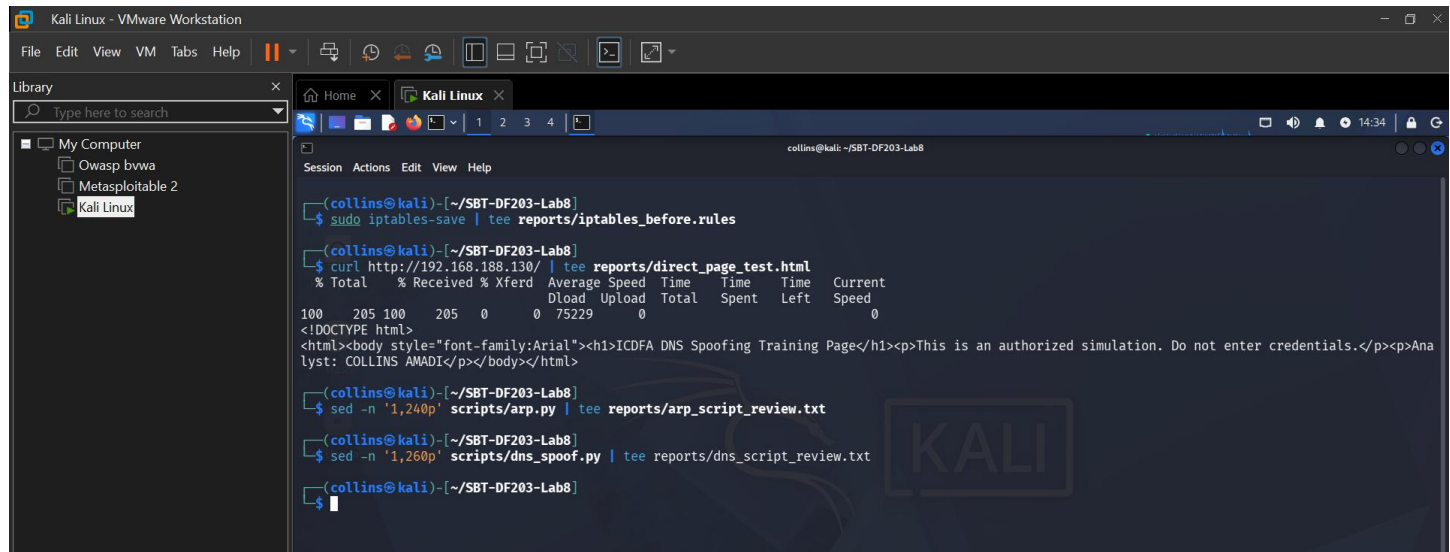
```
sysctl net.ipv4.ip_forward | tee reports/ip_forward_before.txt
sudo iptables-save | tee reports/iptables_before.rules
curl http://ANALYST_IP/ | tee reports/direct_page_test.html
```

Review script content before execution:

```
sed -n '1,240p' scripts/arp.py | tee reports/arp_script_review.txt
```

```
sed -n '1,260p' scripts/dns_spoof.py | tee reports/dns_script_review.txt
```

ANSWER:



```
(collins@kali)~[/SBT-DF203-Lab8]
$ sudo iptables-save | tee reports/iptables_before.rules

(collins@kali)~[/SBT-DF203-Lab8]
$ curl http://192.168.188.130/ | tee reports/direct_page_test.html
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           %             %             Dload  Upload  Total   Spent    Left   Speed
100    205  100    205    0     0    75229      0      0     0      0      0     0      0
<!DOCTYPE html>
<html><body style="font-family:Arial"><h1>ICDFA DNS Spoofing Training Page</h1><p>This is an authorized simulation. Do not enter credentials.</p><p>Ana
lyst: COLLINS AMADI</p></body></html>

(collins@kali)~[/SBT-DF203-Lab8]
$ sed -n '1,240p' scripts/arp.py | tee reports/arp_script_review.txt

(collins@kali)~[/SBT-DF203-Lab8]
$ sed -n '1,260p' scripts/dns_spoof.py | tee reports/dns_script_review.txt

(collins@kali)~[/SBT-DF203-Lab8]
$
```

10. Part C - Capture the Controlled Spoofing Event

Terminal 1 on analyst VM

```
sudo tshark -i IFACE -f 'host VICTIM_IP and (arp or port 53 or tcp port 80)' \
-a duration:60 -w evidence/dns_spoof_controlled.pcapng
```

Other terminals, only after instructor approval:

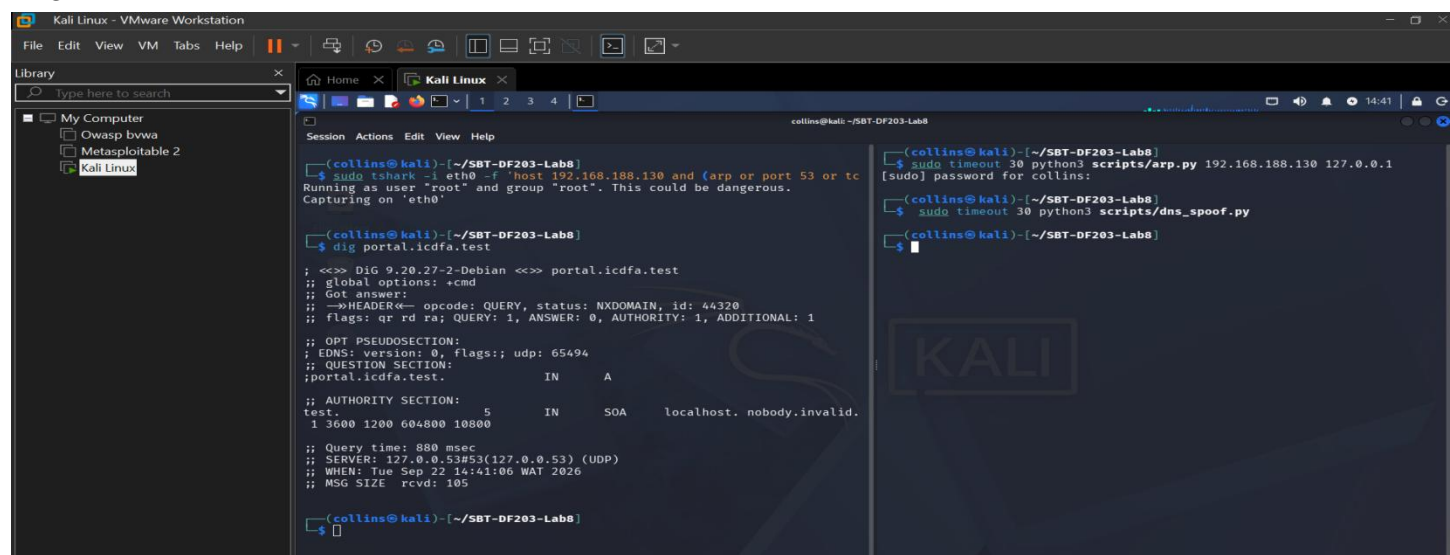
```
# sudo timeout 30 python3 scripts/arp.py VICTIM_IP GATEWAY_IP
```

```
# sudo timeout 30 python3 scripts/dns_spoof.py
```

On victim during the capture:

Clear only the training DNS cache if instructed, then resolve portal.icdfa.test and open it.

ANSWER:



```
(collins@kali)~[/SBT-DF203-Lab8]
$ sudo tshark -i eth0 -f 'host 192.168.188.130 and (arp or port 53 or tc
Running as user "root" and group "root". This could be dangerous.
Capturing on 'eth0'

(collins@kali)~[/SBT-DF203-Lab8]
$ dig portal.icdfa.test

; <<<> DiG 9.20.27-2-Debian <<> portal.icdfa.test
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NXDOMAIN, id: 44320
;; flags: qr rd ra; QUERY: 1, ANSWER: 0, AUTHORITY: 1, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;portal.icdfa.test.          IN      A
;; AUTHORITY SECTION:
test. 5 IN SOA localhost. nobody.invalid.
1 3600 1200 604800 10800

;; Query time: 880 msec
;; SERVER: 127.0.0.53(127.0.0.53) (UDP)
;; WHEN: Tue Sep 22 14:41:06 WAT 2026
;; MSG SIZE rcvd: 105

(collins@kali)~[/SBT-DF203-Lab8]
$

(collins@kali)~[/SBT-DF203-Lab8]
$ sudo timeout 30 python3 scripts/arp.py 192.168.188.130 127.0.0.1
[sudo] password for collins:

(collins@kali)~[/SBT-DF203-Lab8]
$ sudo timeout 30 python3 scripts/dns_spoof.py

(collins@kali)~[/SBT-DF203-Lab8]
$
```

11. Part D - Detect DNS Spoofing Indicators

PCAP=evidence/dns_spoof_controlled.pcapng

sha256sum "\$PCAP" | tee reports/dns_spoof_capture_sha256.txt

DNS queries and all responses

```
tshark -r "$PCAP" -Y 'dns' -T fields \
-e frame.number -e frame.time_epoch -e eth.src -e eth.dst -e ip.src -e udp.srcport \
-e ip.dst -e udp.dstport -e dns.id -e dns.flags.response -e dns.qry.name -e dns.qry.type \
-e dns.flags.rcode -e dns.a -e dns.resp.ttl \
| tee reports/dns_all_fields.tsv
```

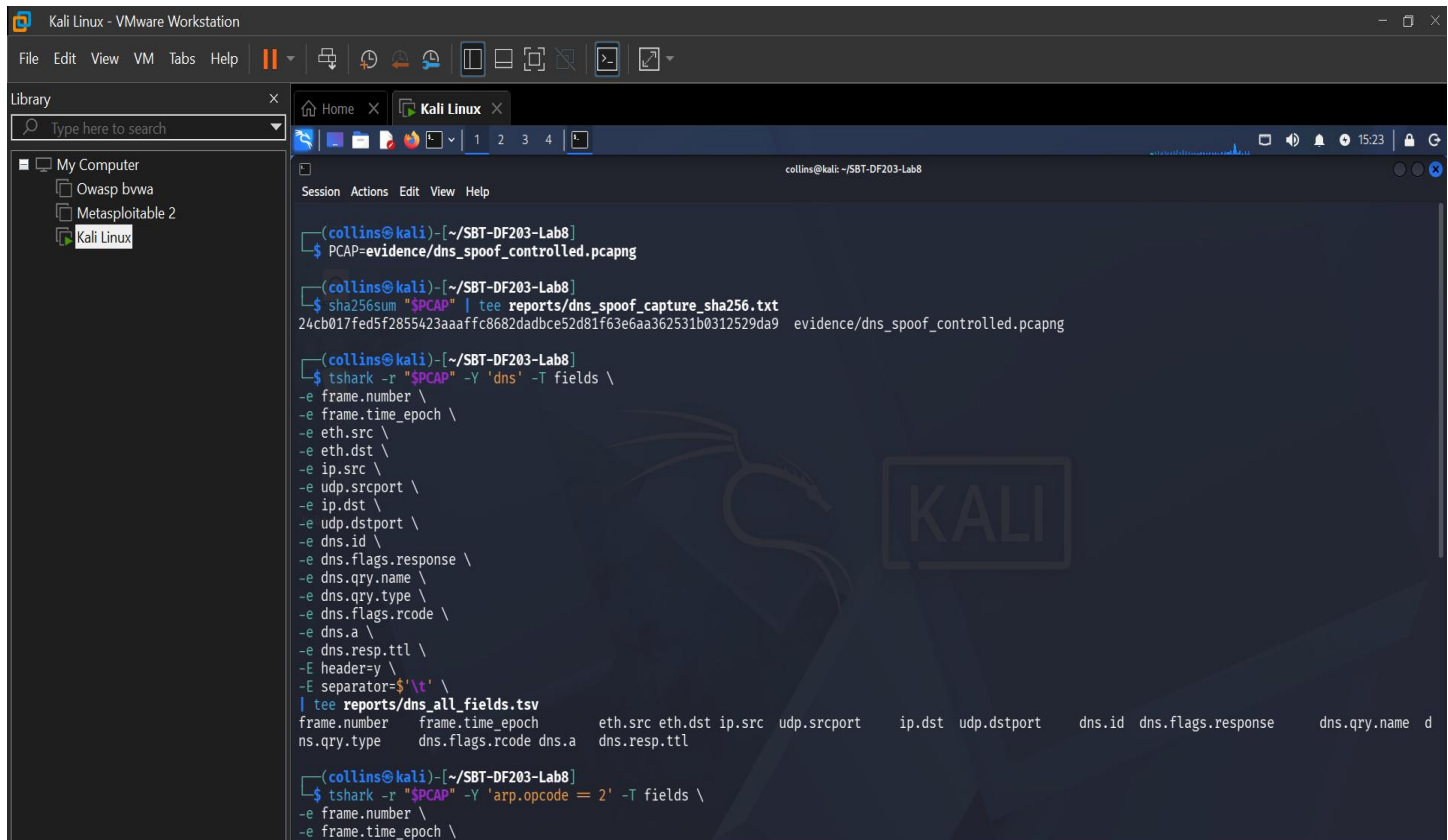
ARP claims involving victim or gateway

```
tshark -r "$PCAP" -Y 'arp.opcode==2' -T fields \
-e frame.number -e frame.time_epoch -e arp.src.proto_ipv4 -e arp.src.hw_mac \
-e arp.dst.proto_ipv4 -e arp.dst.hw_mac \
| tee reports/arp_claims_during_spoof.tsv
```

Subsequent HTTP connection

```
tshark -r "$PCAP" -Y 'http.request || tcp.flags.syn==1' -T fields \
-e frame.number -e frame.time_epoch -e ip.src -e tcp.srcport -e ip.dst -e tcp.dstport -e http.host -e
http.request.uri \
| tee reports/post_dns_connections.tsv
```

ANSWER:



```
Kali Linux - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
Owasp bwwa
Metasploitable 2
Kali Linux
Kali Linux
collins@kali: ~/SBT-DF203-Lab8
Session Actions Edit View Help
(collins@kali)-[~/SBT-DF203-Lab8]
$ PCAP=evidence/dns_spoof_controlled.pcapng
(collins@kali)-[~/SBT-DF203-Lab8]
$ sha256sum "$PCAP" | tee reports/dns_spoof_capture_sha256.txt
24cb017fed5f2855423aaaffc8682dadfce52d81f63e6aa362531b0312529da9 evidence/dns_spoof_controlled.pcapng
(collins@kali)-[~/SBT-DF203-Lab8]
$ tshark -r "$PCAP" -Y 'dns' -T fields \
-e frame.number \
-e frame.time_epoch \
-e eth.src \
-e eth.dst \
-e ip.src \
-e ip.dst \
-e udp.srcport \
-e udp.dstport \
-e dns.id \
-e dns.flags.response \
-e dns.qry.name \
-e dns.qry.type \
-e dns.flags.rcode \
-e dns.a \
-e dns.resp.ttl \
-E header=y \
-E separator='<tab>' \
| tee reports/dns_all_fields.tsv
frame.number      frame.time_epoch  eth.src  eth.dst  ip.src  udp.srcport  ip.dst  udp.dstport  dns.id  dns.flags.response  dns.qry.name  d
ns.qry.type      dns.flags.rcode  dns.a    dns.resp.ttl
(collins@kali)-[~/SBT-DF203-Lab8]
$ tshark -r "$PCAP" -Y 'arp.opcode == 2' -T fields \
-e frame.number \
-e frame.time_epoch \
```

```

-e arp.src.proto_ipv4 \
-e arp.src.hw_mac \
-e arp.dst.proto_ipv4 \
-e arp.dst.hw_mac \
-E header=y \
-E separator='\t' \
| tee reports/arp_claims_during_spoof.tsv
frame.number      frame.time_epoch  arp.src.proto_ipv4  arp.src.hw_mac  arp.dst.proto_ipv4  arp.dst.hw_mac

(collins@kali)~[~/SBT-DF203-Lab8]
$ tshark -r "$PCAP" \
-Y 'http.request || tcp.flags.syn == 1' \
-T fields \
-e frame.number \
-e frame.time_epoch \
-e ip.src \
-e tcp.srcport \
-e ip.dst \
-e tcp.dstport \
-e http.host \
-e http.request.uri \
-E header=y \
-E separator='\t' \
| tee reports/post_dns_connections.tsv
frame.number      frame.time_epoch  ip.src  tcp.srcport  ip.dst  tcp.dstport  http.host  http.request.uri

(collins@kali)~[~/SBT-DF203-Lab8]
$

```

To return to your computer, move the mouse pointer outside or press Ctrl+Alt.

12. Part E - Baseline-versus-Spoof Comparison

Indicator	Baseline	Controlled Spoof	Assessment
DNS responder IP			
DNS responder MAC			
Transaction ID match	65494	44320	
Answer IP			
TTL			
Response time		14:41:06	
Competing answers		NXDOMAIN	
Gateway ARP mapping			
Subsequent HTTP destination			

- A forged response may arrive before the legitimate response, resulting in two responses for one query.
- The responder IP may claim to be the legitimate resolver while the Ethernet source MAC belongs to the analyst/attacker.
- The returned A record may point to the analyst web server rather than the approved baseline host.
- ARP replies may show the gateway or victim IP mapped to the analyst MAC.
- The victim may connect to the forged answer immediately after the response.

13. Part F - Cleanup and Verification

Stop scripts first

```
sudo pkill -f 'arp.py' 2>/dev/null || true
```

```
sudo pkill -f 'dns_spoof.py' 2>/dev/null || true
```

Restore forwarding to its recorded value; example below disables it

```
sudo sysctl -w net.ipv4.ip_forward=0
```

Remove only lab-created iptables/NFQUEUE rules or restore the saved lab snapshot

```
sudo iptables -L -n -v --line-numbers | tee reports/iptables_after_cleanup.txt
```

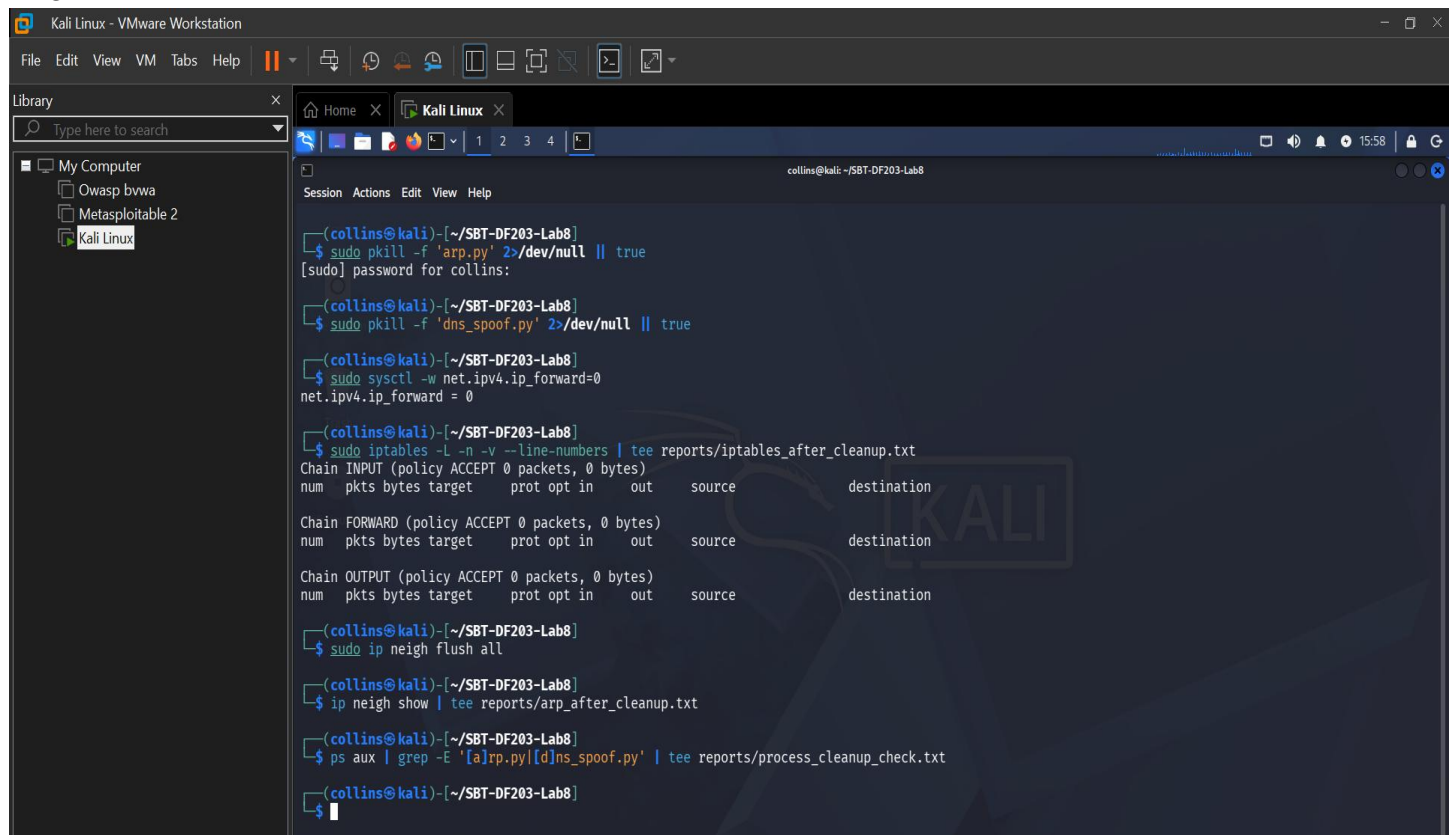
Clear learned ARP entries on the isolated VMs and reconnect interfaces

```
sudo ip neigh flush all
```

```
ip neigh show | tee reports/arp_after_cleanup.txt
```

```
ps aux | grep -E '[a]rp.py|[d]ns_spoof.py' | tee reports/process_cleanup_check.txt
```

ANSWER:



```
Kali Linux - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
Owasp bwba
Metasploitable 2
Kali Linux

(collins@kali)~[/SBT-DF203-Lab8]
$ sudo pkill -f 'arp.py' 2>/dev/null || true
[sudo] password for collins:

(collins@kali)~[/SBT-DF203-Lab8]
$ sudo pkill -f 'dns_spoof.py' 2>/dev/null || true

(collins@kali)~[/SBT-DF203-Lab8]
$ sudo sysctl -w net.ipv4.ip_forward=0
net.ipv4.ip_forward = 0

(collins@kali)~[/SBT-DF203-Lab8]
$ sudo iptables -L -n -v --line-numbers | tee reports/iptables_after_cleanup.txt
Chain INPUT (policy ACCEPT 0 packets, 0 bytes)
num  pkts bytes target    prot opt in     out     source                   destination
Chain FORWARD (policy ACCEPT 0 packets, 0 bytes)
num  pkts bytes target    prot opt in     out     source                   destination
Chain OUTPUT (policy ACCEPT 0 packets, 0 bytes)
num  pkts bytes target    prot opt in     out     source                   destination

(collins@kali)~[/SBT-DF203-Lab8]
$ sudo ip neigh flush all

(collins@kali)~[/SBT-DF203-Lab8]
$ ip neigh show | tee reports/arp_after_cleanup.txt

(collins@kali)~[/SBT-DF203-Lab8]
$ ps aux | grep -E '[a]rp.py|[d]ns_spoof.py' | tee reports/process_cleanup_check.txt

(collins@kali)~[/SBT-DF203-Lab8]
$
```

14. Detection and Defensive Recommendations

- Monitor DNS responses from unexpected MAC addresses or hosts.
- Detect multiple answers for the same transaction with conflicting IPs.
- Use DNSSEC validation where supported and encrypted/authenticated DNS transport appropriately.
- Deploy Dynamic ARP Inspection, DHCP snooping, switch port security and network segmentation.

- Use HTTPS certificate validation; a forged DNS answer should not produce a valid certificate for the real service.
- Correlate DNS, ARP, endpoint cache, switch, web proxy and TLS evidence.

Required Screenshots Checklist

- ☐ Harmless training page and hash.
- ☐ Baseline ARP table and DNS response.
- ☐ Reviewed script target/domain settings.
- ☐ Controlled capture hash.
- ☐ DNS query and all response records.
- ☐ Conflicting or forged DNS answer.
- ☐ ARP claims during event.
- ☐ Victim HTTP connection to replacement IP.
- ☐ Baseline-versus-spoof table.
- ☐ Cleanup, restored ARP and no remaining processes.

Student Report Template

9. Executive summary and authorization.
10. Lab network diagram.
11. Baseline evidence and hashes.
12. Controlled simulation method.
13. DNS anomaly timeline.
14. ARP correlation.
15. Victim connection correlation.
16. Alternative explanations and confidence level.
17. Cleanup and defensive recommendations.
18. Appendix.

Submission Requirements

- One PDF report named SBT-DF203-Lab8_RegNo_FullName.pdf.
- The original or generated PCAP/PCAPNG evidence file and its SHA-256 hash text file.

- A reports folder containing exported CSV/TXT command output and any recovered objects.
- Screenshots must be readable, numbered and referenced in the report narrative.
- Compress the report and evidence outputs into one ZIP named with your registration number and lab number.
- Do not submit passwords or decoded credentials in public discussion channels; include them only in the protected assessment submission where required.

Marking Rubric (100 Marks)

Assessment Area	Marks	What the Instructor Will Check
Safety and preparation	20	Reserved domain, no credential capture, host-only network and baseline recorded.
DNS analysis	25	Correct query/response, transaction, responder, answer and TTL analysis.
ARP and connection correlation	20	Links DNS anomaly to ARP and subsequent traffic.
Forensic interpretation	15	Considers competing explanations and states confidence.
Cleanup and defenses	10	Restored environment and appropriate controls.
Report quality	10	Professional timeline and screenshots.

Instructor Quick Answer Guide

Checkpoint	Expected Observation
Core indicator	DNS answer points to an unauthorized replacement IP.
Strong packet indicator	Resolver IP appears with an unexpected Ethernet source MAC.
Race indicator	Two conflicting responses share one transaction/query.
ARP correlation	Gateway/victim mapping changes to analyst MAC.
Impact evidence	Victim connects to the forged answer.
HTTPS control	Certificate validation limits successful impersonation of real sites.

Command Reference Summary

Command	Purpose
dns.flags.response	Query/response indicator.
dns.id	Match transaction.
dns.a	IPv4 answer.
dns.resp.ttl	Response TTL.
arp.src.proto_ipv4 / hw_mac	ARP IP-to-MAC claim.
http.host	Requested HTTP host.

Academic Integrity and Authorization Statement

Student Declaration

I confirm that I completed this lab in an authorized environment, preserved the supplied evidence, did not target any third-party system, and accurately documented my own commands, observations and conclusions.